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Logic Dungeon I

THE HALL OF INSCRIPTIONS

Purpose: Use the logic tools from Weeks 1–3 to decide which door is best supported.

The Three Doors

Twenty adventurers enter the first underground passage toward *Castrum Malae Ratiocinationis*. Their lamps are failing. To continue, they must find the passage most likely to rise toward natural daylight. Three doors face them:

- **Laurel Door** — carved with leaves; a brass lantern burns beside it.
- **Echo Door** — hung with a bell; its first chamber is dry and uncracked.
- **Slate Door** — marked with a square; pale sand lies beneath its frame.

Every station has two jobs. First, solve the logic problem. Second, decide what the expedition may record about the doors. A clue may be true yet useless. A bad argument may contain a useful fact. A later clue may make an earlier observation important.

Rule of evidence: *Use only the supplied evidence. Record what it establishes, what it does not establish, and whether it helps distinguish among the doors. The goal is the best-supported next step, not a guarantee of safety.*

Tools available: conclusion, stated reason, hidden assumption, follow-test; stable terms; subject and predicate; quality and quantity; A/E/I/O; scope; limits of logic.

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Part 1: The Laurel Chorus — Danger Station

Above the Laurel Door, one hundred small bronze hands are fixed to the wall. When the lantern warms them, they clap. An inscription reads:

“Travelers have applauded the Laurel passage for a hundred years. Therefore Laurel is the wisest road to daylight.”

Question 1.1

Identify the conclusion, stated reason, and hidden assumption. Decide whether the conclusion follows.

Part 1 Q1

Question 1.2

What may the expedition record about Laurel after solving Q1? State what the applause establishes, what it does not establish, and whether this clue helps choose the door.

Part 1 Q2

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Part 2: The Two-Voiced Guide — Danger Station

A stone guide beside the Echo Door announces:

“The Echo passage is sound; masons found no cracks in its first chamber. Every sound choice deserves trust. Therefore Echo is the sound road to daylight.”

Question 2.1

Identify the word that changes meaning. Explain both meanings, rewrite the argument with one stable meaning, and decide what follows.

Part 2 Q1

Question 2.2

What may the expedition record about Echo? Separate the useful fact hidden inside the bad argument from what remains unproved. Does it help with safety, direction, both, or neither?

Part 2 Q2

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Part 3: The Three-Chamber Field Ledger

A scout sampled only the first chamber behind each door. The ledger is **not** a complete census of any passage.

Laurel

Two ash moths circling the brass lantern; both winged.

Echo

Two cave beetles beside the dry wall; one winged and one not winged.

Slate

Two glass ants crossing the pale sand; both winged and moving in a single line beneath the square mark.

The scout writes:

- A. All observed Laurel ash moths are winged creatures.
- B. No observed Echo cave beetles are winged creatures.
- C. Some observed Slate glass ants are winged creatures.
- D. Most creatures in the Hall are winged creatures.

Question 3.1

For A, B, and C, identify A/E/I/O form, quality, and quantity. For C, identify subject and predicate. Which claim is contradicted by the ledger?

Part 3 Q1

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Question 3.2

Which claim reaches farthest beyond the sample and is least useful for choosing a door? Explain. Then write one precise route observation the expedition may preserve for later.

Part 3 Q2

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Part 4: The Naturalist's Key

Farther along the hall, the expedition finds a recent field card stamped by the Academy naturalist:

1. *Ash moths circle any warm lamp, whether or not natural daylight is nearby.*
2. *Cave beetles live in both upper and lower tunnels.*
3. *All glass-ant trail lines carrying pale sand lead toward openings reached by natural daylight.*
4. *Some glass ants wander without forming a trail line.*

Question 4.1

Put Statement 3 into clear A form. Identify its subject, predicate, quality, and quantity. Explain why Statement 4 does not contradict it.

Part 4 Q1

Question 4.2

Combine the key with the earlier ledger. Which observations are now useful, useless, or distracting for the search for daylight? Explain the strongest route inference without changing *some* into *all*.

Part 4 Q2

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Part 5: The Compass and the Old Map

Two final objects lie on a plinth:

The compass: *Its needle points toward Laurel. A small plate beneath it says, “Iron hinge directly behind Laurel Door.”*

The mapmaker’s oath: *“I copied every line carefully from an older map. Therefore this map gives the true condition of all three passages today.” The old map has no date or maker.*

Question 5.1

Use the four-step test on the mapmaker’s oath. Then explain what logic can decide and what requires observation, testimony, or corroboration.

Part 5 Q1

Question 5.2

Evaluate the compass as route evidence. Is its direction false, true but irrelevant, or useful? Explain what the plate allows the expedition to infer.

Part 5 Q2

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Part 6: Expedition Evidence Chart

Complete the chart before choosing. Use short phrases. Do not write “safe” unless the evidence establishes safety.

Door	Established evidence	Still unproved	Help toward day-light?
Laurel			
Echo			
Slate			

Question 6.1

Beneath the chart, identify the strongest clue, one useful-but-limited clue, and one distractor. Explain each classification briefly.

Part 6 Q1

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Part 7: The Door Decision

Question 7.1

Which door is the best-supported next step toward daylight? Use at least three pieces of evidence from different parts. Explain why each rejected door has weaker directional support and state what remains unknown about your chosen passage.

Part 7 Q1

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Part 8: What Logic Has Not Given Us

The company turns toward one door. Before anyone opens it, the expedition leader asks for one final act of precision.

Question 8.1

Name the next observation that would most improve the decision. Then complete: “Logic helped us decide _____, but it could not by itself establish _____.”

Part 8 Q1

Question 8.2

A nervous adventurer says, “Because Slate is best supported, Slate must be completely safe.” Identify the conclusion, hidden assumption, and error. Repair the claim.

Part 8 Q2